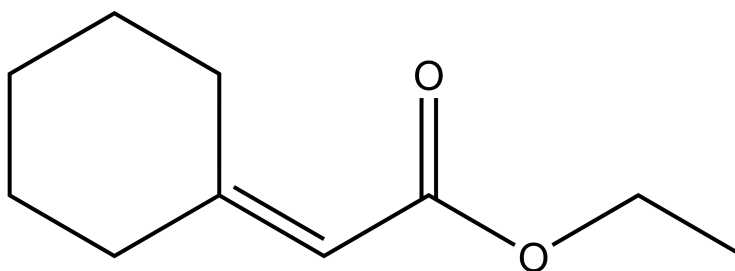


---

# The Synthesis of Ethyl 2-Cyclohexylideneacetate by the Wittig (HWE) Reaction

---



Loganathan Visva [vloganathan@student.ethz.ch](mailto:vloganathan@student.ethz.ch)  
Interdisciplinary Sciences Biochemical-Physical Major

Teaching Assistant  
Neves Vieira Adriana [adriana.vieira@org.chem.ethz.ch](mailto:adriana.vieira@org.chem.ethz.ch)

---

**Abstract** The objective of this experiment is to synthesize Ethyl 2-cyclohexylideneacetate and analyze the yield of the product. The synthesis is performed through the Wittig (HWE) reaction of Triethyl phosphonoacetate with cyclohexanone and the usage of magnesium chloride and triethylamine as catalyst. The substance was purified by flash column chromatography. The success of the experiment is determined by thin layer chromatography and infrared spectrum measurements. The product was a pale yellowish clear liquid with a weight of 1.04 g, what would be a yield of about 61%.

---

Zürich, April 5, 2024

  
Loganathan Visva

## Introduction

Ethyl 2-cyclohexylideneacetate is an organic compound characterized by the presence of a cyclohexylidene group bonded to an ester functional group. At room temperature, it appears as a colorless to pale yellow liquid. It has a characteristic odor that smells fruity or like floral scents [1]. This compound's structure features a cyclohexene ring conjugated with an ester, which contributes to its chemical reactivity and solubility characteristics. It is not typically soluble in water but shows good solubility in organic solvents like dichloromethane and ethylacetate [2].

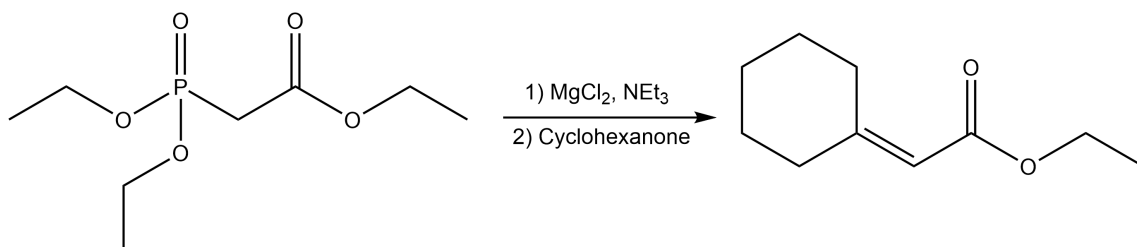
In industrial applications, Ethyl 2-cyclohexylideneacetate is primarily utilized as an intermediate in pharmaceutical manufacturing [3]. Its chemical properties make it suitable for this role, where it is involved in the synthesis of various medicinal products. Its unique odor profile could also be utilized in the formulation of fragrances, scents and perfumes.

## Material Properties

In the described experiment, Ethyl 2-cyclohexylideneacetate is produced through the Horner-Wadsworth-Emmons (HWE) reaction.

This is a modified version of the Wittig reaction, that utilized phosphonate-stabilized carbanions [4]. These carbanions, in comparison to ylides in the Wittig reaction, exhibit greater nucleophilicity and reduced basicity, allowing for effective alkylation. Ylides are a class of organic compounds that contain a positively charged atom (usually phosphorus, sulfur, or nitrogen) directly bonded to a negatively charged carbon atom. This results in a neutral molecule that has both positive and negative charges, which are separated by a covalent bond. Additionally, the byproduct formed in this reaction, a dialkylphosphate salt, is readily separable from the desired product via simple aqueous extraction.

This synthesis involves reacting triethylphosphonoacetate with cyclohexanone, utilizing triethylamine as a base to facilitate the reaction.



| Substance                                       | GHS Hazard Pictogram                                                                 | H and P Phrases                                                                                                                                                                            |
|-------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cyclohexanone <a href="#">[5]</a>               |     | H226, H302, H332, H312, H315, H318, H335-P210, P280, P301, P312, P303, P361, P353, P304, P340, P305, P351, P338                                                                            |
| Triethyl phosphonoacetate <a href="#">[6]</a>   |     | H319, H411-P264, P273, P280, P305, P351, P338, P337, P313, P391                                                                                                                            |
| Triethylamine <a href="#">[7]</a>               |    | H225, H302, H331, H311, H314, H318, H335-P210, P280, P301, P312, P303, P361, P353, P304, P340, P310, P305, P351, P338                                                                      |
| Tetrahydrofuran <a href="#">[8]</a>             |     | H225, H302, H319, H351, H336, H335-P202, P210, P233, P301, P312, P305, P351, P338, P308, P313                                                                                              |
| Hydrochloric acid <a href="#">[9]</a>           |  | H225, H290, H315, H319, H350, H335, H314, H318-P202, P210, P233, P303, P361, P353, P305, P351, P338, P308, P313                                                                            |
| Ethyl ether <a href="#">[10]</a>                |   | H224, H302, H336-P210, P233, P240, P241, P301, P312, P403                                                                                                                                  |
| Ethylcyclohexylideneacetate <a href="#">[1]</a> |   | H302, H315, H319, H335, P261, P264, P264+P265, P270, P271, P280, P301+P317, P302+P352, P304+P340, P305+P351+P338, P319, P321, P330, P332+P317, P337+P317, P362+P364, P403+P233, P405, P501 |
| Magnesium chloride <a href="#">[11]</a>         | -                                                                                    | -                                                                                                                                                                                          |
| Sodium chloride <a href="#">[12]</a>            | -                                                                                    | -                                                                                                                                                                                          |
| Magnesium sulfate <a href="#">[13]</a>          | -                                                                                    | -                                                                                                                                                                                          |

| Substance                                     | Molar Mass<br>[g/mol]    | Density<br>[g/mL]      | Melting Point<br>[°C]                | Boiling Point<br>[°C]                |
|-----------------------------------------------|--------------------------|------------------------|--------------------------------------|--------------------------------------|
| Cyclohexanone <a href="#">5</a>               | 98.15                    | 0.9478                 | -47                                  | 155.65                               |
| Triethyl phosphonoacetate <a href="#">6</a>   | 224.19                   | 1.0917                 | 92-93                                | 261                                  |
| Triethylamine <a href="#">7</a>               | 101.19                   | 0.7255                 | -114.7                               | 88.8                                 |
| Tetrahydrofuran <a href="#">8</a>             | 72.107                   | 0.8876                 | -108.4                               | 66                                   |
| Hydrochloric acid <a href="#">9</a>           | 36.46                    | 1.09620                | -114                                 | -85                                  |
| Ethyl ether <a href="#">10</a>                | 74.12                    | 0.713                  | -116.3                               | 34.6                                 |
| Ethylcyclohexylideneacetate <a href="#">1</a> | 168.23                   | 1.0                    | 6                                    | 137                                  |
| Magnesium chloride <a href="#">11</a>         | 95.211                   | 2.32                   | 714                                  | 1,412                                |
| Sodium chloride <a href="#">12</a>            | 58.443                   | 2.17                   | 800.7                                | 1,465                                |
| Magnesium sulfate <a href="#">13</a>          | 120.366<br>(anhydrous)   | 2.66<br>(anhydrous)    | 1,124<br>(anhydrous,<br>decomposes)  | 1,124<br>(anhydrous,<br>decomposes)  |
|                                               | 246.47<br>(heptahydrate) | 1.68<br>(heptahydrate) | 150<br>(heptahydrate,<br>decomposes) | 150<br>(heptahydrate,<br>decomposes) |

## Safety Assessment

In the lab, each chemical demands specific safety measures. Cyclohexanone and triethylamine are flammable, with triethylamine also being corrosive. Tetrahydrofuran is not only flammable but forms hazardous peroxides. Hydrochloric acid can cause severe burns, and both it and ethyl ether are harmful when inhaled. Magnesium chloride and sodium chloride are less hazardous but can irritate the eyes.

Also the synthesis should be conducted in a fume hood, because of the harmful vapors. Triethylamine should be added slowly to the mixture because it warms up and that shouldn't be too drastic. Stir the mixture for at least 10 minutes before adding the cyclohexanone drop wise.

The handling of each requires appropriate personal protective equipment, cautious storage, and clear protocols for exposure response and spill management. Always adhere to detailed guidelines as provided by their respective Safety Data Sheets for safe laboratory practices.

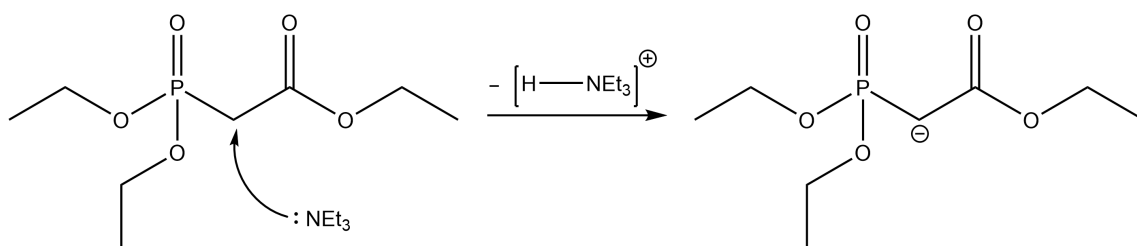
For avoiding any health issues that could be caused by the chemicals, gloves and safety goggles should be worn in the laboratory. In a case of an accident the emergency number should be called. With the internal ETH-phones it would be 888, on personal mobile phones the number would be 044 342 11 88. If something gets into the eye, an eye shower has to be performed for at least 15 minutes and then the eye clinic from the university hospital Zurich has to be visited. If sulfuric acid gets spilled on the skin, the clothing over it has to be removed and the contaminated area on the skin has to be rinsed with running water for several minutes. Afterwards a doctor must be seen. All incidents should always be reported.

## Waste Disposal

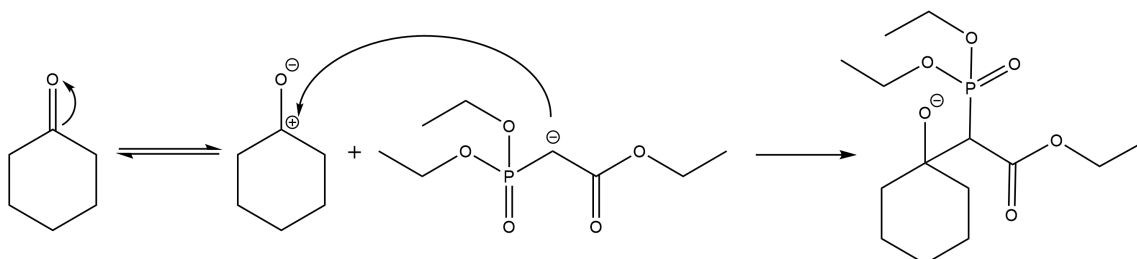
Hydrochloric acid should be disposed into the aqueous acidic waste. Cyclohexanone, ethylether, tetrahydrofuran and triethylamine are organic solvents and should be disposed in the solvent waste. Magnesium chloride and Sodium chloride should be either disposed in the solid waste or dissolved and put in the chlorinated solvent waste. Magnesium sulfate should also be put in the solid waste or dissolved and put in the solvent waste.

## Reaction Mechanism

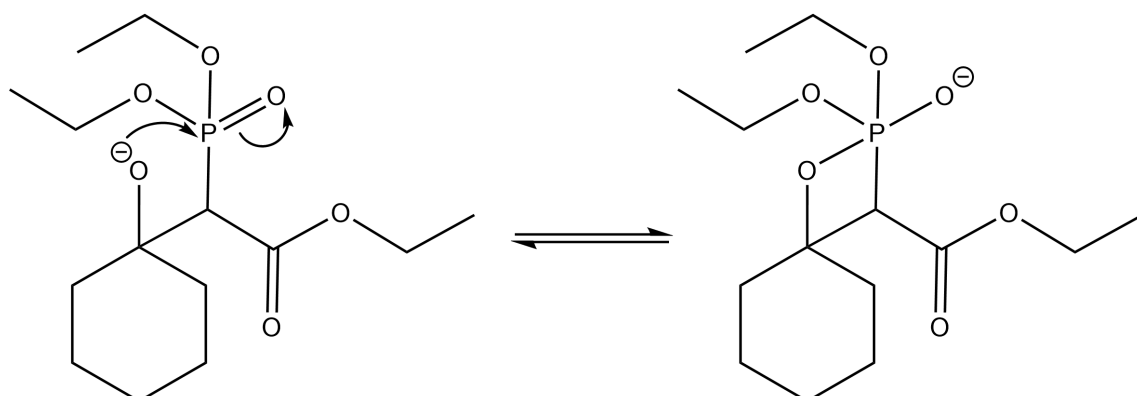
In this synthesis, the HWE reaction, which is an modification of the Wittig reaction, is utilised. Initially, the triethylamine acts as a base to deprotonate triethylphosphonoacetate. This action results in a carbanion that is negatively charged.



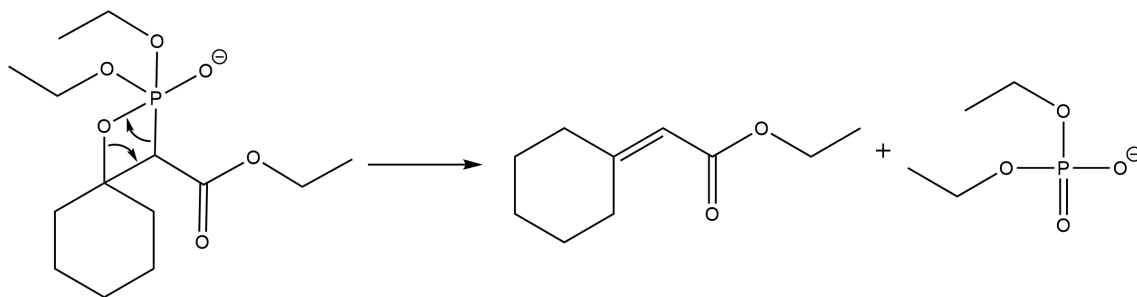
The carbanion is a nucleophile that attacks the electrophilic carbonyl group from the cyclohexanone. That results in the connection of both molecules.



The double bond's electrons between the phosphorus and oxygen atoms shift towards the oxygen, leading to a positive charge on the phosphorus. This allows a nucleophilic attack by the negatively charged oxygen atom of the cyclohexanone molecule.



The Bayer ring strain is really high on the quadratic ring with the phosphorus (P), the oxygen (O) and the two carbon (C) atoms. That leads to the redistribution of the electrons from the C-P and the C-O bonds. That enables the formation of a C-C double bond and a P-O double bond. This rearrangement results in the release of the end product ethyl 2-cyclohexylideneacetate and the byproduct diethylphosphate.



## Atom Economy

Atom economy refers to the efficiency of a chemical reaction and it is calculated with the equation below. The atom economy of the reaction is approximately 52%. This reaction is not really atom efficient.

$$AE = \frac{M_{\text{Product}}}{M_{\text{Reagents}}} = \frac{168.23 \text{ g/mol}}{224.19 \text{ g/mol} + 98.15 \text{ g/mol}} \approx 52.19\%$$

## Experimental

### Procedure

Ethyl 2-cyclohexylideneacetate is produced through the Horner-Wadsworth-Emmons (HWE) reaction. The synthesis process utilizes triethylphosphonoacetate and cyclohexanone as reactants, with triethylamine is utilized as a basic catalyst. The product was a pale yellowish clear liquid (1.04 g, 6.182 mmol) with a yield of about 61%.

### Observation

For the synthesis of the ethyl 2-cyclohexylideneacetate, a magnetic stir bar was placed in a 100 mL three-neck flask filled with anhydrous magnesium chloride (10 mmol, 0.952g, 1.00 equiv.). A nitrogen supply was connected to one neck of the flask, a dropping funnel was installed on another neck and on the third neck a septum was put on. With a nitrogen atmosphere was created inside the three-necked flask. That was necessary because in this synthesis tetrahydrofuran was utilized. THF should be kept under a nitrogen atmosphere to prevent peroxide formation. THF can autoxidize upon exposure to air, especially oxygen, leading to the accumulation of dangerous peroxides, which can be explosive. A syringe was utilized to insert the tetrahydrofuran (10 ml) and triethyl phosphonoacetate (10 mmol, 2.1 mL, 1.00 equiv.) through the septum. The mixture was stirred for 5 minutes at room temperature. Next, triethylamine (10 mmol, 1.4 mL, 1.00 equiv.) was slowly added with a syringe through the septum. After an additional 10 minutes of stirring at room temperature, cyclohexanone (10 mmol, 1.0 mL, 1.00 equiv.) was dripped slowly into the mixture in the same way. The mixture became a thick, viscous, milky-white liquid after some time. That happened because the nitrogen flow was too high, what caused the evaporation of the THF, so additional 10 mL of THF were added. That caused the milky-white solution to become thin-liquid again. After that the nitrogen flow was stopped and the whole system was sealed, all potential escape points for the nitrogen were securely closed up. The resultant mixture was then stirred continuously at room temperature for a duration of 20 hours under the nitrogen atmosphere.

After 20 hours, 10 mL of a 2 M hydrochloric acid were added slowly into the mixture. Then the mixture was transferred into a separation funnel and shaken. A organic phase with lower density accumulated on top, lower down a aqueous phase accumulated. Both phases were separated into flasks. The aqueous phase was transferred into the separation funnel again and 10 mL of ethylacetate were added and the mixture was shaken. That caused again a separate organic phase on top of a aqueous one. The organic phase was collected in the erlenmeyer flask. This process was performed three times. The combined organic layers were washed with a brine solution. The organic phase was subsequently dried over magnesium sulfate and filtered through a 50 mL filter crucible. The residue was disposed, the ethyl 2-cyclohexylidene was inside the mother liquors but still with

solvent. With the help of a rotary evaporator the solvent got removed and a white crystalline solid remained.

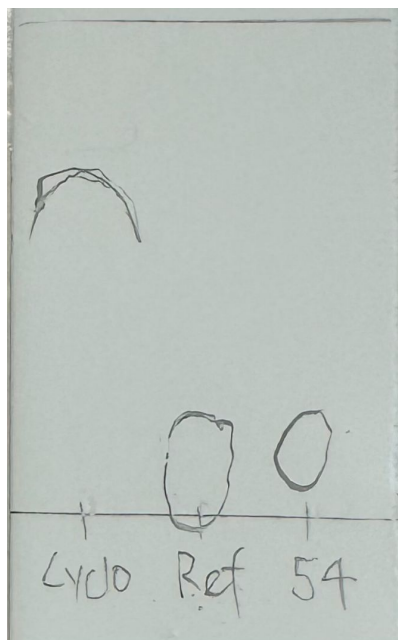
A chromatography column was filled up with a sand layer, a silica layer and a sand layer again. Then the crystalline solid was then put on top of the sand layer. Then on top of the crystalline solid another sand layer was put on top. A 20 to 1 n-Hexane ethylacetate mixture was produced. This mixture was transferred into the chromatography column. The column was connected to the nitrogen flow, so pressure could be applied to it. Then a flash column chromatography was performed and after that the solution containing the desired product was put in a rotatory evaporator.

## Characterisation

### TLC and Flash Column Chromatography

A flash column chromatography was conducted to purify the 2-cyclohexylidene acetate. A flash column chromatography is just like a longer reverse thin layer chromatography for the entire mixture. In whole there were 54 flasks that were filled. Some TLC's were conducted, that showed that there was no product inside flask 1 to 11. Flask 1 to 21 had the product. Flask 22-26 had some impurity, maybe a byproduct. Flask 27 to flask 54 again only had the product. So the content of flask 1 to 21 and flask 27 to 54 were put together into a 1 L round bottomed flask. It appeared as a clear liquid.

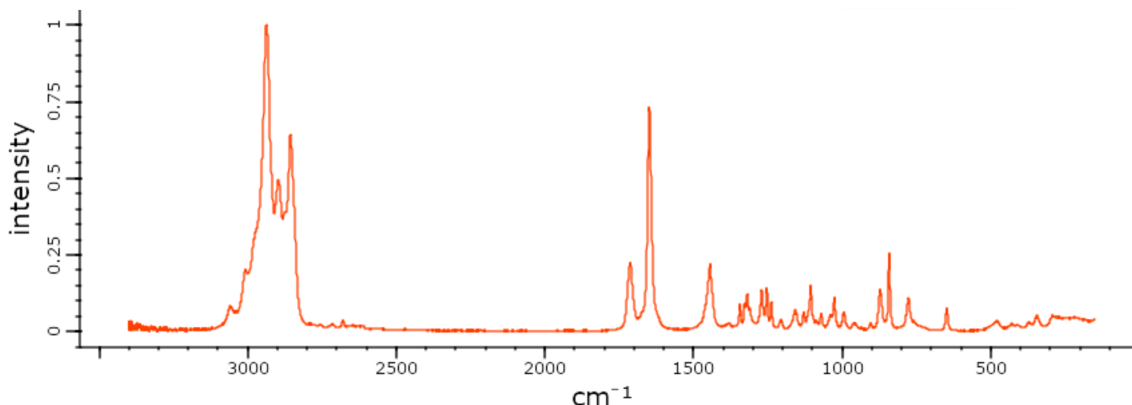
A final TLC was conducted with the liquid in the round bottomed flask. In this TLC, a silicate plate was used and there were three columns. The first column (Cyclo) had only one educt, the cyclohexanone. For the second column (Ref) n-Hexane was put into the flask where the white crystalline substance was originally to take up the residues from the substance. This acted as a reference. The third column (54) only had some drops from flask 54 on it. The solvent consisted of 100% n-Hexane.



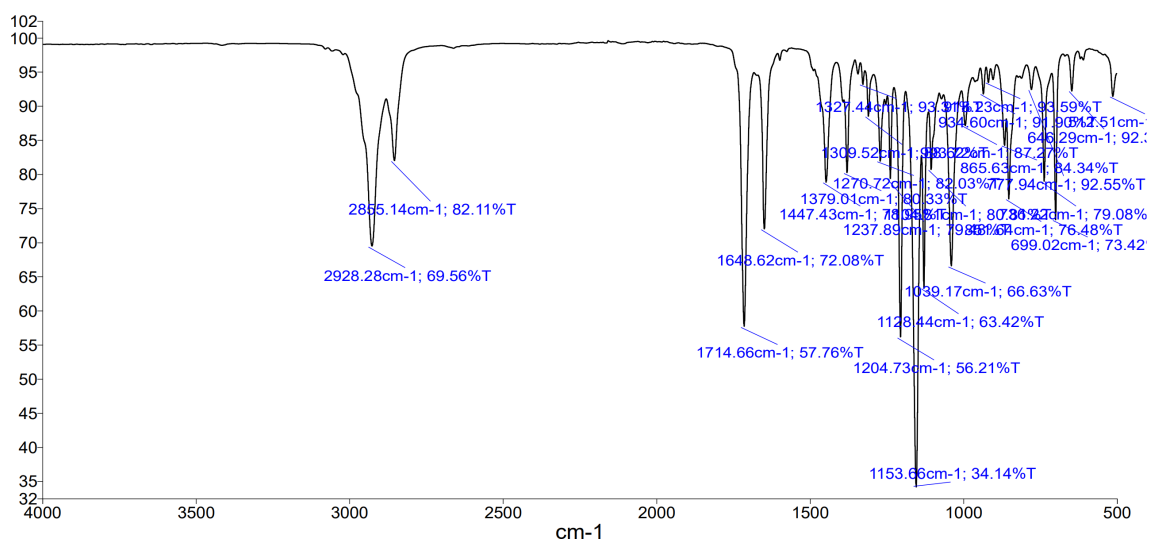
This TLC plate shows that there is no more cyclohexanone in the white crystalline solid and also not in flask 54. The  $R_f$  values are 0.66 for cyclohexanone, 0.1 for the white crystalline solid and 0.14 for flask 54.

## IR Spectrum

This would be the literature values of the infrared spectrum of ethyl 2-cyclohexylidene acetate [14].



And this would be the measured infrared spectrum of the synthesized ethyl 2-cyclohexylidene acetate from the author.



In the spectral analysis, the observed peaks align with those documented in the literature, although they are weaker in intensity [15]. The wavelengths are accurate, but the absorption rates are lower, indicated by the T%, potentially due to the presence of impurities, variations in sample concentration, or the influence of the instrument's sensitivity and configuration.



| Wavenumber (cm <sup>-1</sup> ) | Bond                              | %T    |
|--------------------------------|-----------------------------------|-------|
| 2855.14                        | C-H Stretch (Alkane)              | 82.11 |
| 2928.28                        | C-H Stretch (Alkane)              | 69.56 |
| 1714.66                        | C=O Stretch (Carbonyl)            | 57.76 |
| 1648.62                        | C=C Stretch (Alkene)              | 72.08 |
| 1447.43                        | C-H Bending                       | 78.95 |
| 1379.01                        | C-H Bending                       | 80.33 |
| 1327.44                        | C-H Bending                       | 93.31 |
| 1237.89                        | C-H Bending                       | 79.48 |
| 1204.73                        | C-O Stretch (Ester)               | 56.21 |
| 1153.66                        | C-O Stretch (Ester/Ether/Alcohol) | 34.14 |
| 1128.44                        | C-O Stretch (Aliphatic Ether)     | 63.42 |
| 1039.17                        | Unknown/Impurity                  | 66.63 |

## References

- [1] Ethyl cyclohexylideneacetate. <https://pubchem.ncbi.nlm.nih.gov/compound/Ethyl-cyclohexylideneacetate> Accessed: 2024-03-05.
- [2] Ethyl cyclohexylideneacetate. <https://www.fishersci.ch/shop/products/ethyl-cyclohexylideneacetate-98-thermo-scientific/11476057> Accessed: 2024-03-05.
- [3] Ethyl cyclohexylideneacetate, 98 <https://www.thermofisher.com/order/catalog/product/B25073.22> Accessed: 2024-03-05.
- [4] Dave Farina. Horner-wadsworth-emmons reaction. [https://youtu.be/5lIw\\_MhirBM?si=hNBwibIrDl20q806](https://youtu.be/5lIw_MhirBM?si=hNBwibIrDl20q806)
- [5] Cyclohexanone. <https://www.matmake.com/s/cyclohexanone> Accessed: 2024-03-04.
- [6] Triethyl phosphonoacetate. [https://en.wikipedia.org/wiki/Triethyl\\_phosphonoacetate](https://en.wikipedia.org/wiki/Triethyl_phosphonoacetate) Accessed: 2024-03-04.
- [7] Triethylamine. <https://www.chemeurope.com/en/encyclopedia/Triethylamine.html> Accessed: 2024-03-04.
- [8] Tetrahydrofuran (thf). <https://www.sciencemadness.org/smwiki/index.php/Tetrahydrofuran> Accessed: 2024-03-04.
- [9] Hydrochloric acid - the definitive guide. <https://www.biologydictionary.net/hydrochloric-acid/> Accessed: 2024-03-04.
- [10] Diethyl ether. [https://en.wikipedia.org/wiki/Diethyl\\_ether](https://en.wikipedia.org/wiki/Diethyl_ether) Accessed: 2024-03-04.
- [11] Magnesium chloride. [https://en.wikipedia.org/wiki/Magnesium\\_chloride](https://en.wikipedia.org/wiki/Magnesium_chloride) Accessed: 2024-03-04.
- [12] Sodium chloride. [https://en.wikipedia.org/wiki/Sodium\\_chloride](https://en.wikipedia.org/wiki/Sodium_chloride) Accessed: 2024-03-04.
- [13] Magnesium sulfate. [https://en.wikipedia.org/wiki/Magnesium\\_sulfate](https://en.wikipedia.org/wiki/Magnesium_sulfate) Accessed: 2024-03-04.
- [14] Wiley-VCH Verlag GmbH Co. KGaA. IR spectrum and properties of ethyl cyclohexylideneacetate. In IR - Industrial Chemicals, Pure Organic Compounds - Wiley, 2024. Accessed on KnowItAllAnywhere on March 31, 2024.
- [15] Infrared spectroscopy absorption table. [https://chem.libretexts.org/Ancillary\\_Materials/Reference/Reference\\_Tables/Spectroscopic\\_Reference\\_Tables/Infrared\\_Spectroscopy\\_Absorption\\_Table](https://chem.libretexts.org/Ancillary_Materials/Reference/Reference_Tables/Spectroscopic_Reference_Tables/Infrared_Spectroscopy_Absorption_Table) Accessed on 2024-02-.

## Appendix

### List of H and P Phrases

- H225: Highly flammable liquid and vapor.
- H302: Harmful if swallowed.
- H312: Harmful in contact with skin.
- H315: Causes skin irritation.
- H318: Causes serious eye damage.
- H319: Causes serious eye irritation.
- H331: Toxic if inhaled.
- H332: Harmful if inhaled.
- H335: May cause respiratory irritation.
- H336: May cause drowsiness or dizziness.
- H411: Toxic to aquatic life with long lasting effects.
- H290: May be corrosive to metals.
- H314: Causes severe skin burns and eye damage.
- H351: Suspected of causing cancer.
- P210: Keep away from heat, sparks, open flames, hot surfaces. — No smoking.
- P233: Keep container tightly closed.
- P240: Ground/bond container and receiving equipment.
- P241: Use explosion-proof electrical/ventilating/lighting equipment.
- P242: Use only non-sparking tools.
- P243: Take precautionary measures against static discharge.
- P264: Wash skin thoroughly after handling.
- P273: Avoid release to the environment.
- P280: Wear protective gloves/protective clothing/eye protection/face protection.
- P301+P312: IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell.
- P303+P361+P353: IF ON SKIN (or hair): Remove/Take off immediately all contaminated clothing. Rinse skin with water/shower.
- P304+P340: IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing.
- P305+P351+P338: IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses if present and easy to do. Continue rinsing.
- P308+P313: IF exposed or concerned: Get medical advice/attention.
- P337+P313: If eye irritation persists: Get medical advice/attention.
- P403+P233: Store in a well-ventilated place. Keep container tightly closed.
- P405: Store locked up.
- P501: Dispose of contents/container in accordance with local/regional/national/international regulations.

## Table from the Experiment Manual

| Reactant                     | MW     | Equiv | Moles   | Mass g | Volume mL | Purity |
|------------------------------|--------|-------|---------|--------|-----------|--------|
| Cyclohexanone                | 98.15  | 1.00  | 10 mmol | 0.9815 | 1.0356    | -      |
| Triethylphosphonoacetate     | 224.19 | 1.00  | 10 mmol | 2.2419 | 2.0550    | -      |
| Triethylamine                | 101.19 | 1.00  | 10 mmol | 1.0119 | 1.3948    | -      |
| Magnesium chloride           | 95.21  | 1.00  | 10 mmol | 0.9521 | -         | -      |
| Product                      | MW     | Yield | Moles   | Mass   |           |        |
| Ethyl-2-Cyclohexylidenacetat | 168.23 |       |         |        |           |        |



## Infrared Spectroscopy Absorption Table

The following table lists **infrared spectroscopy absorptions** by frequency regions.

### 4000-3000 $\text{cm}^{-1}$

|                        |        |       |     |            |                         |                                           |
|------------------------|--------|-------|-----|------------|-------------------------|-------------------------------------------|
| 3700-3584              | medium | sharp | O-H | stretching | alcohol                 | free                                      |
| 3550-3200              | strong | broad | O-H | stretching | alcohol                 | intermolecular bonded                     |
| 3500- 3400             | medium | -     | N-H | stretching | primary amine           | -                                         |
| 3400-3300<br>3330-3250 | medium | -     | N-H | stretching | aliphatic primary amine | -                                         |
| 3350-3310              | medium | -     | N-H | stretching | secondary amine         | -                                         |
| 3300-2500              | strong | broad | O-H | stretching | carboxylic acid         | usually centered on 3000 $\text{cm}^{-1}$ |
| 3200-2700              | weak   | broad | O-H | stretching | alcohol                 | intramolecular bonded                     |
| 3000-2800              | strong | broad | N-H | stretching | amine salt              | -                                         |

### 3000-2500 $\text{cm}^{-1}$

|           |        |       |     |            |          |         |
|-----------|--------|-------|-----|------------|----------|---------|
| 3333-3267 | strong | sharp | C-H | stretching | alkyne   | -       |
| 3100-3000 | medium | -     | C-H | stretching | alkene   | -       |
| 3000-2840 | medium | -     | C-H | stretching | alkane   | -       |
| 2830-2695 | medium | -     | C-H | stretching | aldehyde | doublet |
| 2600-2550 | weak   | -     | S-H | stretching | thiol    | -       |

### 2400-2000 $\text{cm}^{-1}$

|           |        |       |       |            |                |                 |
|-----------|--------|-------|-------|------------|----------------|-----------------|
| 2349      | strong | -     | O=C=O | stretching | carbon dioxide | -               |
| 2275-2250 | strong | broad | N=C=O | stretching | isocyanate     | -               |
| 2260-2222 | weak   | -     | C≡N   | stretching | nitrile        | -               |
| 2260-2190 | weak   | -     | C≡C   | stretching | alkyne         | disubstituted   |
| 2175-2140 | strong | -     | S-C≡N | stretching | thiocyanate    | -               |
| 2160-2120 | strong | -     | N=N=N | stretching | azide          | -               |
| 2150      | -      | -     | C=C=O | stretching | ketene         | -               |
| 2145-2120 | strong | -     | N=C=N | stretching | carbodiimide   | -               |
| 2140-2100 | weak   | -     | C≡C   | stretching | alkyne         | monosubstituted |
| 2140-1990 | strong | -     | N=C=S | stretching | isothiocyanate | -               |
| 2000-1900 | medium | -     | C=C=C | stretching | allene         | -               |
| 2000      | -      | -     | C=C=N | stretching | ketenimine     | -               |

### 2000-1650 $\text{cm}^{-1}$

|  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|
|  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|



|           |      |   |     |         |                   |          |
|-----------|------|---|-----|---------|-------------------|----------|
| 2000-1650 | weak | - | C-H | bending | aromatic compound | overtone |
|-----------|------|---|-----|---------|-------------------|----------|

1870-1540  $\text{cm}^{-1}$ 

|              |        |   |     |            |                                   |                                          |
|--------------|--------|---|-----|------------|-----------------------------------|------------------------------------------|
| 1818<br>1750 | strong | - | C=O | stretching | anhydride                         | -                                        |
| 1815-1785    | strong | - | C=O | stretching | acid halide                       | -                                        |
| 1800-1770    | strong | - | C=O | stretching | conjugated acid halide            | -                                        |
| 1775<br>1720 | strong | - | C=O | stretching | conjugated anhydride              | -                                        |
| 1770-1780    | strong | - | C=O | stretching | vinyl / phenyl ester              | -                                        |
| 1760         | strong | - | C=O | stretching | carboxylic acid                   | monomer                                  |
| 1750-1735    | strong | - | C=O | stretching | esters                            | 6-membered lactone                       |
| 1750-1735    | strong | - | C=O | stretching | $\delta$ -lactone                 | $\gamma$ : 1770                          |
| 1745         | strong | - | C=O | stretching | cyclopentanone                    | -                                        |
| 1740-1720    | strong | - | C=O | stretching | aldehyde                          | -                                        |
| 1730-1715    | strong | - | C=O | stretching | $\alpha,\beta$ -unsaturated ester | or formates                              |
| 1725-1705    | strong | - | C=O | stretching | aliphatic ketone                  | or cyclohexanone or cyclopentanone       |
| 1720-1706    | strong | - | C=O | stretching | carboxylic acid                   | dimer                                    |
| 1710-1680    | strong | - | C=O | stretching | conjugated acid                   | dimer                                    |
| 1710-1685    | strong | - | C=O | stretching | conjugated aldehyde               | -                                        |
| 1690         | strong | - | C=O | stretching | primary amide                     | free (associated: 1650)                  |
| 1690-1640    | medium | - | C=N | stretching | imine / oxime                     | -                                        |
| 1685-1666    | strong | - | C=O | stretching | conjugated ketone                 | -                                        |
| 1680         | strong | - | C=O | stretching | secondary amide                   | free (associated: 1640)                  |
| 1680         | strong | - | C=O | stretching | tertiary amide                    | free (associated: 1630)                  |
| 1650         | strong | - | C=O | stretching | $\delta$ -lactam                  | $\gamma$ : 1750-1700 $\beta$ : 1760-1730 |

1670-1600  $\text{cm}^{-1}$ 

|  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|
|  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|



|           |        |   |     |            |                                    |                       |
|-----------|--------|---|-----|------------|------------------------------------|-----------------------|
| 1678-1668 | weak   | - | C=C | stretching | alkene                             | disubstituted (trans) |
| 1675-1665 | weak   | - | C=C | stretching | alkene                             | trisubstituted        |
| 1675-1665 | weak   | - | C=C | stretching | alkene                             | tetrasubstituted      |
| 1662-1626 | medium | - | C=C | stretching | alkene                             | disubstituted (cis)   |
| 1658-1648 | medium | - | C=C | stretching | alkene                             | vinylidene            |
| 1650-1600 | medium | - | C=C | stretching | conjugated alkene                  | -                     |
| 1650-1580 | medium | - | N-H | bending    | amine                              | -                     |
| 1650-1566 | medium | - | C=C | stretching | cyclic alkene                      | -                     |
| 1648-1638 | strong | - | C=C | stretching | alkene                             | monosubstituted       |
| 1620-1610 | strong | - | C=C | stretching | $\alpha,\beta$ -unsaturated ketone | -                     |

1600-1300  $\text{cm}^{-1}$ 

|                        |        |   |     |            |                |                 |
|------------------------|--------|---|-----|------------|----------------|-----------------|
| 1550-1500<br>1372-1290 | strong | - | N-O | stretching | nitro compound | -               |
| 1465                   | medium | - | C-H | bending    | alkane         | methylene group |
| 1450<br>1375           | medium | - | C-H | bending    | alkane         | methyl group    |
| 1390-1380              | medium | - | C-H | bending    | aldehyde       | -               |
| 1385-1380<br>1370-1365 | medium | - | C-H | bending    | alkane         | gem dimethyl    |

1400-1000  $\text{cm}^{-1}$ 

|                        |        |   |     |            |                   |   |
|------------------------|--------|---|-----|------------|-------------------|---|
| 1440-1395              | medium | - | O-H | bending    | carboxylic acid   | - |
| 1420-1330              | medium | - | O-H | bending    | alcohol           | - |
| 1415-1380<br>1200-1185 | strong | - | S=O | stretching | sulfate           | - |
| 1410-1380<br>1204-1177 | strong | - | S=O | stretching | sulfonyl chloride | - |
| 1400-1000              | strong | - | C-F | stretching | fluoro compound   | - |
| 1390-1310              | medium | - | O-H | bending    | phenol            | - |
| 1372-1335<br>1195-1168 | strong | - | S=O | stretching | sulfonate         | - |
| 1370-1335<br>1170-1155 | strong | - | S=O | stretching | sulfonamide       | - |



|                        |        |       |         |            |                   |                              |
|------------------------|--------|-------|---------|------------|-------------------|------------------------------|
| 1350-1342<br>1165-1150 | strong | -     | S=O     | stretching | sulfonic acid     | anhydrous hydrate: 1230-1120 |
| 1350-1300<br>1160-1120 | strong | -     | S=O     | stretching | sulfone           | -                            |
| 1342-1266              | strong | -     | C-N     | stretching | aromatic amine    | -                            |
| 1310-1250              | strong | -     | C-O     | stretching | aromatic ester    | -                            |
| 1275-1200<br>1075-1020 | strong | -     | C-O     | stretching | alkyl aryl ether  | -                            |
| 1250-1020              | medium | -     | C-N     | stretching | amine             | -                            |
| 1225-1200<br>1075-1020 | strong | -     | C-O     | stretching | vinyl ether       | -                            |
| 1210-1163              | strong | -     | C-O     | stretching | ester             | -                            |
| 1205-1124              | strong | -     | C-O     | stretching | tertiary alcohol  | -                            |
| 1150-1085              | strong | -     | C-O     | stretching | aliphatic ether   | -                            |
| 1124-1087              | strong | -     | C-O     | stretching | secondary alcohol | -                            |
| 1085-1050              | strong | -     | C-O     | stretching | primary alcohol   | -                            |
| 1070-1030              | strong | -     | S=O     | stretching | sulfoxide         | -                            |
| 1050-1040              | strong | broad | CO-O-CO | stretching | anhydride         | -                            |

1000-650 cm<sup>-1</sup>

|                    |        |   |      |            |               |                       |
|--------------------|--------|---|------|------------|---------------|-----------------------|
| 995-985<br>915-905 | strong | - | C=C  | bending    | alkene        | monosubstituted       |
| 980-960            | strong | - | C=C  | bending    | alkene        | disubstituted (trans) |
| 895-885            | strong | - | C=C  | bending    | alkene        | vinylidene            |
| 850-550            | strong | - | C-Cl | stretching | halo compound | -                     |
| 840-790            | medium | - | C=C  | bending    | alkene        | trisubstituted        |
| 730-665            | strong | - | C=C  | bending    | alkene        | disubstituted (cis)   |
| 690-515            | strong | - | C-Br | stretching | halo compound | -                     |
| 600-500            | strong | - | C-I  | stretching | halo compound | -                     |

900-700 cm<sup>-1</sup>

|                                    |        |   |     |         |                      |   |
|------------------------------------|--------|---|-----|---------|----------------------|---|
| 880 ± 20<br>810 ± 20               | strong | - | C-H | bending | 1,2,4-trisubstituted | - |
| 880 ± 20<br>780 ± 20<br>(700 ± 20) | strong | - | C-H | bending | 1,3-disubstituted    | - |





|                        |        |   |     |         |                                               |   |
|------------------------|--------|---|-----|---------|-----------------------------------------------|---|
| 810 ± 20               | strong | - | C-H | bending | 1,4-disubstituted or 1,2,3,4-tetrasubstituted | - |
| 780 ± 20<br>(700 ± 20) | strong | - | C-H | bending | 1,2,3-trisubstituted                          | - |
| 755 ± 20               | strong | - | C-H | bending | 1,2-disubstituted                             | - |
| 750 ± 20<br>700 ± 20   | strong | - | C-H | bending | monosubstituted benzene derivative            | - |

### Contributors and Attributions

- [OChemOnline](#)

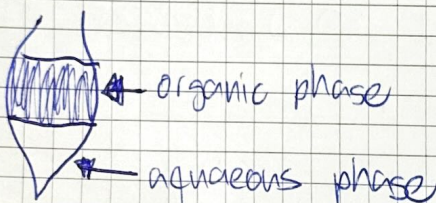
Infrared Spectroscopy Absorption Table is shared under a [CC BY-NC-SA 4.0](#) license and was authored, remixed, and/or curated by LibreTexts.

| BOOK                                                                                                     |  | PAGE |  | PROJECT                 |  |
|----------------------------------------------------------------------------------------------------------|--|------|--|-------------------------|--|
|                                                                                                          |  | - 8  |  |                         |  |
| TITLE                                                                                                    |  |      |  | PROJECT                 |  |
| Continued from page                                                                                      |  |      |  |                         |  |
| <u>Exp. 3a</u>                                                                                           |  |      |  |                         |  |
| <u>Safety</u>                                                                                            |  |      |  |                         |  |
| 3-Ethylphosphate acetate                                                                                 |  |      |  |                         |  |
| Cyclohexanone                                                                                            |  |      |  |                         |  |
| Tetrahydrofuran                                                                                          |  |      |  |                         |  |
| Don't open the nitrogen valves too quickly + over pressure adapts<br>→ Can happen with light and oxygen! |  |      |  |                         |  |
| Diethylether → Can build peroxides / highly volatile<br>→ Don't sniff them!                              |  |      |  |                         |  |
| Triethylamin smells like fish                                                                            |  |      |  |                         |  |
| → 3-neck flask } - septum<br>} - nitrogen supply<br>} - dropping funnel                                  |  |      |  |                         |  |
| <hr/>                                                                                                    |  |      |  |                         |  |
| MgCl <sub>2</sub> 0,952g                                                                                 |  |      |  |                         |  |
| THF 10mL                                                                                                 |  |      |  |                         |  |
| Triethyl Phosphonacetate 2,05mL                                                                          |  |      |  |                         |  |
| 5 min stir                                                                                               |  |      |  |                         |  |
| Triethylamin 1,4 mL                                                                                      |  |      |  |                         |  |
| 10 min stir                                                                                              |  |      |  |                         |  |
| SIGNATURE                                                                                                |  |      |  | DATE                    |  |
| DISCLOSED TO AND UNDERSTOOD BY                                                                           |  |      |  | DATE                    |  |
|                                                                                                          |  |      |  | PROPRIETARY INFORMATION |  |
| Continued to page                                                                                        |  |      |  |                         |  |



| TITLE                                                                                                                                                                                                                                                                                                                                                                       |  | PROJECT |                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---------|-------------------------|
| Continued from page                                                                                                                                                                                                                                                                                                                                                         |  | - 9     |                         |
|                                                                                                                                                                                                                                                                                                                                                                             |  | BOOK    | PAGE                    |
| <p>+10mL THF (the previous THF evaporated)<br/>because the nitrogen flow was too high)</p> <p>Cyclohexanone 1,03 mL</p> <p><u>Experimental</u></p> <p><u>Procedure</u></p> <p>Really short description + <sup>+ mass</sup> Ausbeute / yield + colour<br/>of compound + state (liquid/solid)</p> <p><u>Observation</u></p> <p>Describe everything and explain everything</p> |  |         |                         |
| SIGNATURE                                                                                                                                                                                                                                                                                                                                                                   |  | DATE    |                         |
| DISCLOSED TO AND UNDERSTOOD BY                                                                                                                                                                                                                                                                                                                                              |  | DATE    | PROPRIETARY INFORMATION |

Continued to page

| - 10                                                                               |         |
|------------------------------------------------------------------------------------|---------|
| BOOK                                                                               | PAGE    |
| TITLE                                                                              | PROJECT |
| Continued from page                                                                |         |
| After 20h                                                                          |         |
| 10mL 2M HCL                                                                        |         |
| <del>Ethylacetate</del>                                                            |         |
| <u>Separation funnel</u>                                                           |         |
|  |         |
| 10mL x 3mL Ethylacetate in aqueous phase                                           |         |
| Sodium chloride + organic phase                                                    |         |
| Dry it over $MgSO_4$ and filter it                                                 |         |
| 100% n-Hexane TLC                                                                  |         |
| Continued to page                                                                  |         |
| SIGNATURE                                                                          |         |
| DATE                                                                               |         |
| DISCLOSED TO AND UNDERSTOOD BY                                                     |         |
| DATE                                                                               |         |
| PROPRIETARY INFORMATION                                                            |         |



- 11  
BOOK PAGE

**TITLE** **PROJECT**

Continued from page

Column Chromatography  
TLC done multiple times

5  
10  
15  
20  
25  
30  
35  
40  
45

Sand  
Silica → 45g  
Sand  
n-Hexane sagging

TLC  
Column

the smaller the silica - "Schicht"  
the faster the flash column chromatography but also the dirtier the product

the highest one will come out the first!

1L n-Hexane + 50mL Ethylacetate

Continued to page

|                                |  |                         |
|--------------------------------|--|-------------------------|
| SIGNATURE                      |  | DATE                    |
| DISCLOSED TO AND UNDERSTOOD BY |  | DATE                    |
|                                |  | PROPRIETARY INFORMATION |

|                     |              | - 12        |          |
|---------------------|--------------|-------------|----------|
|                     |              | BOOK        | PAGE     |
| TITLE               |              | PROJECT     |          |
| Continued from page |              |             |          |
|                     | Flask        | 12 - 21     |          |
| 5                   | Flask        | 27 - End    |          |
| 10                  | Flask weight | with liquid | 293.65 g |